Can Chance Explain the Mean Difference?

AP Statistics · Unit 4 · Topic 4.2 · B: 2027-01-26 · E: 2027-01-27 · TEACHER KEY

CED TETHER

- **VAR-1.I** Identify questions suggested by probabilities of errors in statistical inference. **[Skill 1.A]**
- **VAR-1.I.1** Random variation may result in errors in statistical inference.

Essential question: How can rerandomization distinguish a treatment explanation from chance variation while preserving the possibility of inferential error?

DOK-3 skill: 4.B · **FRQ pattern:** `rerandomization-probability-error-and-scope`

DOK rationale: Part (c) adjudicates competing causal claims by coordinating the observed mean difference, a rerandomization tail proportion, its conditional meaning, the remaining chance-error risk, and the distinct roles of random assignment and volunteer recruitment.

First take → rules + (a)–(c) → turn in **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Display the board and ask for one sentence using the study description.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board at minute 33. Students finish the shortened parts and justify their judgment.
- Collect at the door. Score part (c) only using the three required elements.

Students do

- Write a one-sentence first take before any discussion.
- Work (a) and (b) from the rules box and the stem.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

Adult role

Circulate and ask for evidence. Accept a concise response; do not require omitted calculations or a second full inference write-up.

[WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **uses-simulation** (required) — Supports Nia using 18/2,000 under no effect
- **allows-error** (required) — Explains that rare under no effect does not make an error impossible
- **limits-scope** (required) — Limits the causal claim to the volunteers rather than all district students

E	Uses the conditional simulation evidence, allows an error, and limits the causal claim to the volunteers.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] TODAY'S PROBLEM — annotated

Suppose 50 student volunteers test a planning prompt for a logic task. They are randomly assigned to two groups of 25. Mean times are 14.2 minutes with the prompt and 17.8 with standard directions: prompt minus standard is -3.6. Under no effect, a simulation shuffles the same times into new groups. Only 18 of 2,000 shuffled differences are -3.6 or lower.

Nia: "This is evidence that the prompt lowered mean time for these volunteers, but we could still be wrong."

Owen: "The 0.009 is the probability of no effect, so there is a 99.1% chance the prompt works for every district student."

Experiment results

Group	n	Mean (min)
Prompt	25	14.2
Standard	25	17.8

FIRST TAKE (5 min)

Whose claim sounds more believable, Nia's or Owen's? Give one reason from the study.
Key: Ungraded. Everyday language is enough before the discussion; formal vocabulary belongs in the finish.

Rules callout "ASK WHAT CHANCE WOULD DO UNDER NO EFFECT (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** What difference in group means would we expect if the prompt had no effect?

Key: The expected prompt-minus-standard difference is 0 minutes.

DOK 2 **(b)** Compute 18/2,000. In one sentence, explain what this proportion estimates if the prompt has no effect.

Key: $18/2{,}000 = 0.009 = 0.9\%$. If the prompt has no effect, about 0.9% of random assignments give a difference of -3.6 minutes or lower.

DOK 3 **(c)** In 3–4 sentences, decide whose claim is better supported. Use the simulation result, explain why an error is still possible, and say why the volunteers do not represent every district student.

Key: Nia is better supported: a difference of -3.6 or lower occurred in only 18 of 2,000 no-effect trials. This estimates the chance of such a result assuming no effect, not the probability that no effect is true, so an incorrect effect claim remains possible. Random assignment supports a causal conclusion for these volunteers. Because they volunteered, the study does not justify a claim about every district student.

EXIT

One thing the rules box changed about my first take: ______________________